

SESSION OBJECTIVES

THE STUDENT PHYSICIANS WILL BE ABLE TO:

1. **Demonstrate** the components of the **history and physical examination** of the elbow and shoulder
2. **Evaluate** the patient utilizing **inspection, palpation, active range of motion, passive range of motion, strength testing** of the elbow and shoulder
3. **Apply** special testing of :the shoulder including **painful arc test, the empty can test, drop-arm test, Yergason test, Apprehension test, Hawkin's test, Neer's test and Apley scratch test**
:and the elbow **including tests for medial and lateral epicondylitis, laxity of the elbow and ulnar nerve irritation**
4. **Distinguish normal versus abnormal examination** findings.
5. Properly **document** normal and abnormal findings of the shoulder and elbow examination in the objective component of the **SOAP note**.

Clinical Decision Making

- How is MSK history different than all the rest?
 - Prior injuries?
 - Repetitive movements at home or work/hobbies and habits
 - Patient may not know the etiology
- After asking your history questions...
 - Time to think about what your potential diagnosis list will look like
- How will this list look after you perform your physical exam?
 - Will things be ruled in or excluded?
 - What other data will you need from the exam and special tests?

MSK History-Where

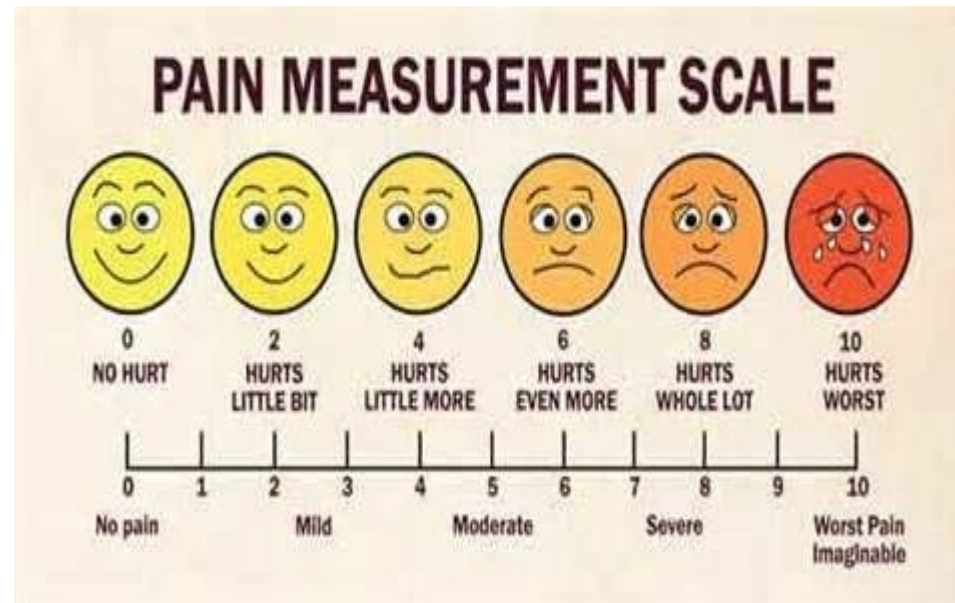
- Articular or Extraarticular
- Point with one finger
- Deep or superficial
- Pinpoint or diffuse
- Mono, Oligo or Polyarticular
- Is there migration from joint to joint
- Intermittent or constant
- Symmetrical or asymmetrical
- Does the pain radiate (hip to groin)

MSK History Quality

- Describe the pain. Initially don't offer options. It is important to allow the patient to use their own words.
- If the patient is having difficulty describing the pain, then you can offer suggestions.
- The quality of the pain in MSK is not the most helpful information but should still be gathered.

MSK History Severity

- Dig a little deeper with MSK
- Initial severity, current severity, worst severity, usual severity
- VAS: visual analog scale is based on the patients report of pain. We most often use a scale of 1-10, but what does that mean? Is your 5 =to my 5?



ASSOCIATED SYMPTOMS-LOCAL

- Is there local swelling, heat, redness,
- Is there numbness, pins/needles, local weakness
- Is there a rash, a bruise, a hematoma
- Is there a deformity, a lump, a bump, a lesion
- Is there stiffness, limited movement, painful movement
- Does it hurt only when they move the joint, or even when someone else moves it
- Has activity been hampered by the local symptoms
- Are ADLs being affected: hair combing, brushing teeth, getting dressed

ASSOCIATED SYMPTOMS- CONSTITUTIONAL

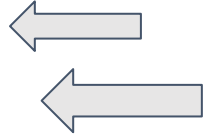
- Fever, chills, night sweats
- Fatigue
- Weight loss,
- Rash
- Generalized weakness

Physical Exam MSK

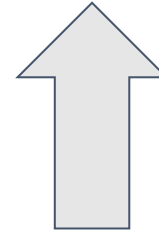
Always look before you touch



ALWAYS EXAMINE BOTH SIDES



Always touch before you move



ALWAYS EXAMINE BOTH SIDES

Ask the patient to move first (AROM) and then you can move the patient (PROM)

NEXT..... Perform strength testing

ALWAYS EXAMINE BOTH SIDES

• **ALWAYS EXAMINE BOTH SIDES**

And finally perform special testing

The MSK Exam always follows the same format

Inspection

Palpation

ROM – Active then passive

MMT

Special testing

Check sensory if there is a neuro component to the patient's complaint

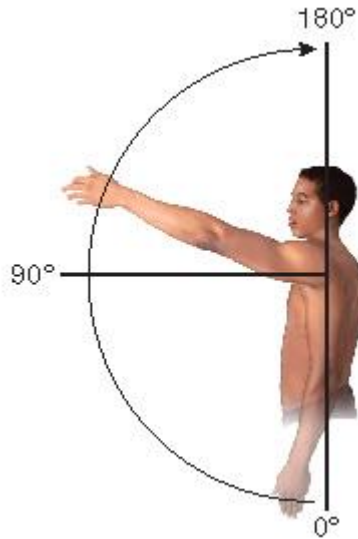
AROM FLEXION

- **Principal Muscles Affecting Movement**

Anterior deltoid, pectoralis major (clavicular head), coracobrachialis, biceps brachii

Patient Instructions

“Raise your arms in front of you and overhead.”





•Internal Rotation

Principal Muscles Affecting Movement

Subscapularis, anterior deltoid, pectoralis major, teres major, latissimus dorsi

Patient Instructions

“Place one hand behind your back and try to touch your shoulder blade.”

Identify the highest midline spinous process the patient is able to reach.



•External Rotation

Principal Muscles Affecting Movement

Infraspinatus, teres minor, posterior deltoid, supraspinatus (especially with arm overhead)

Patient Instructions

“Raise your arm to shoulder level; bend your elbow and rotate your forearm toward the ceiling.”

OR

“Place one hand behind your neck or head as if you are brushing your hair.”

Motor Manual Testing

- MMT should then be performed in all available ranges of motion
- Graded on a scale of 0-5
- 0 (zero) = no visible muscle movement at all
- 1 (trace) = muscle contraction felt but no visible movement of muscle
- 2 (poor) = movement through complete ROM with gravity eliminated
- 3 (fair) = holds test position with no added pressure
- 4 (good) = holds test position against moderate pressure
- 5 (normal) = holds test position against strong pressure

Special Tests

- Crossover or crossed body adduction test -
AC joint



- Apley scratch test –
Shoulder rotation

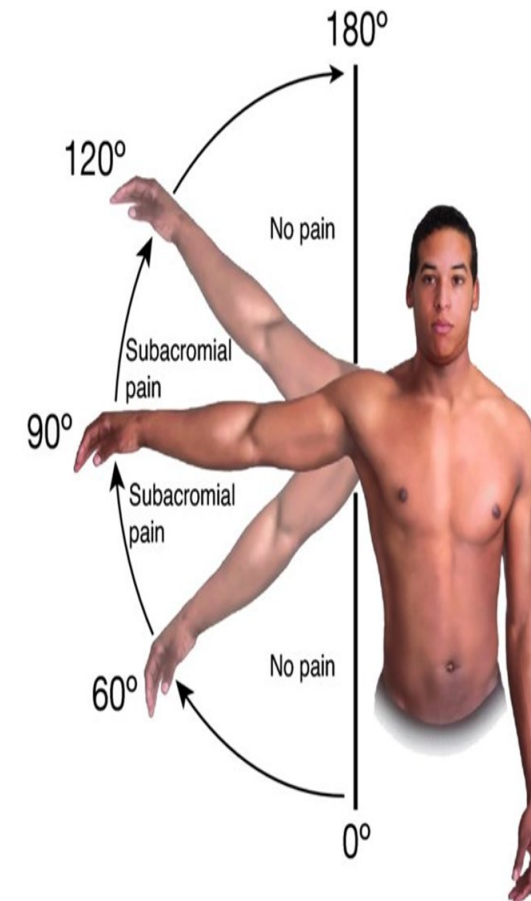


Tests abduction and external rotation.



Tests ADDuction and IR

- Painful arc test-
rotator cuff tendinitis



Bicep Tendinitis

- anterior shoulder pain
- pain is exacerbated by lifting objects and overhead activities
- improves with rest
- localized tenderness over the bicipital groove
- painful with resisted elbow flexion and forearm supination(Yergason)

Shoulder

- Pearls:
 - Cervical pathology may masquerade as shoulder pain(C5)
 - Throwing or racquet sport ask phase (cocking, release, follow through)
 - Posterior dislocation – arm is usually in internal rotation at time of injury
 - Anterior dislocation – arm is usually in external rotation at time of injury



Strength Testing



Elbow Flexion



Elbow Extension

Special Tests

Tinel's of Elbow-testing for cubital tunnel syndrome



Cozen Test-Lateral epicondylitis testing



Medial epicondylitis testing



Elbow

- Pearls
 - Lateral epicondyles (LES)
 - Extensors
 - Supinators
 - Medial epicondyles (MFP)
 - Flexors
 - Pronators
- Do not miss
 - Tendon rupture
 - Dislocation
 - Fractures

How to document the MSK exam

- Which **joint** are you documenting?: you cannot say normal ROM in right upper extremity unless you measured every joint in that arm
- Which **side** are you documenting?: right or left or both
- ROM should be documented as: Patient's finding/normal ROM
- MMT is on a scale from 0-5
- **EX: On examination of the left shoulder Flexion 120/180, abduction 110/180, IR 45/90, ER 70/90, Adduction and extension were normal**
- **MMT revealed 5/5 strength in adduction, extension and 4/5 strength in abduction, flexion, IR and ER.**
- **The right shoulder exam revealed normal ROM and 5/5 strength in all the above movements**